IGCSE 0607 Practice Questions — E1.4 — Combined Questions & Markscheme

Fractions, decimals and percentages

Total Questions: 18 — **Total Marks:** 100 Marks

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1 **(a)** Write the following in order of size, starting with the smallest.

$$0.65 \qquad \frac{2}{3} \qquad 64\% \qquad \frac{5}{8}$$

..... **[2]**

(b) Write 0.85 as a fraction in its simplest form.

..... **[1]**

2 **(a)** Write $0.\dot{7}$ as a fraction.

..... **[1]**

(b) Write $0.\dot{3}\dot{6}$ as a fraction in its simplest form.

..... **[2]**

3 Work out

$$3\frac{1}{3} \div 2\frac{4}{5} - \frac{5}{14}$$

Give your answer as a fraction in its simplest form.

..... **[3]**

4 **(a)** Write 45% as a fraction in its simplest form.

..... **[1]**

(b) Write $\frac{9}{25}$ as a decimal.

..... **[1]**

(c) Write 2.04 as a mixed number in its simplest form.

..... **[2]**

5 **(a)** Write 0.45 as a fraction in its simplest form.

..... **[1]**

(b) Write $\frac{3}{8}$ as a decimal.

..... **[1]**

(c) Write these in order of size, starting with the smallest.

$$\frac{5}{12} \qquad 0.4 \qquad 42\% \qquad \frac{3}{7}$$

..... **[2]**

6 **(a)** Write 0.075 as a fraction in its simplest form.

..... **[2]**

(b) Write $3\frac{2}{5}$ as a decimal.

..... **[1]**

(c) Write the following in order of size, starting with the smallest.

$$62\% \qquad \frac{5}{8} \qquad 0.6 \qquad \frac{7}{11}$$

..... **[2]**

7 The length of a rectangle is increased by $33.\dot{3}\%$.
The width of the rectangle is decreased by $x\%$.
The overall area of the rectangle increases by $16.\dot{6}\%$.
Calculate the value of x.

$x =$ ..... **[5]**

8 **(a)** A tank is initially $\frac{1}{5}$ full of water.
When 24 litres of water are added, the tank is exactly 60% full.
Work out the total capacity of the tank.

..... litres **[3]**

(b) A bottle contains x millilitres of juice.
30% of the juice is poured out.
Then $\frac{1}{4}$ of the remaining juice is poured out.
The bottle now contains 210 millilitres of juice.
Work out the value of x.

$x =$ ..... **[3]**

9 A bag contains red counters, blue counters and green counters.
The number of red counters is x.
The number of blue counters is $2x + 5$.
The number of green counters is $3x - 1$.

(a) Exactly 35% of the counters in the bag are blue.
Work out the value of x.

$x =$ ..... **[4]**

(b) Work out the percentage of the counters in the bag that are green.
Write your answer as a mixed number.

..... % **[3]**

10 **(a)** Write the recurring decimal $0.2\dot{4}\dot{5}$ as a fraction in its simplest form.
You must show all your working.

..... **[3]**

(b) The fraction $\frac{N}{42}$, where N is a positive integer, is in its simplest form.
Explain why the decimal equivalent of $\frac{N}{42}$ is a recurring decimal.

.....

..... **[2]**

(c) The recurring decimal $0.\dot{x}\dot{y}$, where x and y are distinct digits, can be written as a fraction $\frac{k}{11}$, where k is an integer.
Find the value of $x + y$.

$x + y =$ ..... **[2]**

11 **(a)** By converting the recurring decimal to a fraction, show that

$$0.2\dot{7} = \frac{5}{18}$$

[2]

(b) In a school, $\frac{2}{5}$ of the students are boys and the rest are girls.
45% of the boys play a sport.
30% of the girls play a sport.

(i) Work out the percentage of all students in the school who play a sport.

..... % **[3]**

(ii) There are 144 students who play a sport.
Work out the total number of students in the school.

..... **[2]**

12 A train departs a station with x passengers.
At the first stop, $\frac{1}{3}$ of the passengers get off the train and 12 passengers get on.
At the second stop, 25% of the passengers now on the train get off and nobody gets on.
After the second stop, there are 15 fewer passengers on the train than there were when it departed.

(a) Write down an expression in terms of x, in its simplest form, for the number of passengers on the train after the first stop.

..... **[2]**

(b) Form an equation in x and solve it to find the number of passengers on the train when it departed.

$x =$ ..... **[4]**

(c) What percentage of the original passengers remained on the train after the second stop?
Give your answer as a mixed number.

..... % **[3]**

13 In a school of 840 students, 35% are in the lower school, $\frac{3}{8}$ are in the middle school, and the rest are in the upper school.

(a) Work out the number of students in the middle school.

..... **[1]**

(b) Work out the percentage of students in the upper school.

..... % **[2]**

(c) 15% of the middle school students play a musical instrument.
Work out the fraction of the total number of students who are middle school students that play a musical instrument.
Give your answer in its simplest form.

..... **[2]**

14 A tank contains V litres of water.
During the first hour, the tank loses 24% of its water.
During the second hour, the tank loses $\frac{2}{5}$ of its remaining water.
After these two hours, there are 45.6 litres of water left in the tank.

(a) Calculate the value of V.

$V =$ ..... **[4]**

(b) Express the total amount of water lost over the two hours as a fraction of the original volume V.
Give your answer in its simplest form.

..... **[2]**

15 In a school, $\frac{3}{8}$ of the students are in the Junior section, 45% are in the Middle section, and the rest are in the Senior section.
There are 140 students in the Senior section.

(a) Work out the total number of students in the school.

..... **[3]**

(b) The number of boys in the Junior section is 150% of the number of girls in the Junior section.
Work out the number of girls in the Junior section.

..... **[3]**

16 **(a)** Write the following in order of size, starting with the smallest.

$$\frac{5}{7} \qquad 71\% \qquad 0.714 \qquad \frac{3}{4}$$

Smallest ..... **[2]**

(b) Find the fraction which is exactly halfway between $\frac{2}{5}$ and $\frac{5}{8}$.
Give your answer in its simplest form.

..... **[2]**

(c) A fraction is equivalent to $\frac{4}{9}$.
The difference between the denominator and the numerator is 35.
Find the fraction.

..... **[2]**

17 A company has three departments: Sales, Production, and Administration.

- $\frac{2}{5}$ of the employees are in Sales.
- 28% of the employees are in Production.
- The remaining 24 employees are in Administration.

(a) Work out the total number of employees in the company.

..... **[3]**

(b) 60% of the Sales employees are women.
$\frac{4}{7}$ of the Production employees are men.
Half of the Administration employees are women.
Work out the percentage of the whole company who are women.

..... % **[4]**

18 **(a)** A television costs \$450. In a sale, the price is reduced by 18%.
Calculate the sale price of the television.

\$ ..... **[2]**

(b) In the same sale, a laptop is sold for \$629 after a 15% reduction.
Calculate the original price of the laptop.

\$ ..... **[3]**

(c) The price of a tablet is increased by $\frac{1}{5}$.
Later, this new price is decreased by 20%.
Calculate the overall percentage decrease in the price of the tablet.

..... % **[2]**

1 **(a)** $\frac{5}{8}$ 64% 0.65 $\frac{2}{3}$ **[B2]**

M1 for correct conversions to decimals seen: 0.666 . . . , 0.64, 0.625

(b) $\frac{17}{20}$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $\frac{7}{9}$ **[B1]**

(b) $\frac{36}{99}$ **[M1]**

$\frac{4}{11}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 $\frac{10}{3} \div \frac{14}{5}$ **[M1]**

$\frac{10}{3} \times \frac{5}{14}$ or $\frac{25}{21}$ **[M1]**

$\frac{5}{6}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.36 **[B1]**

(c) $\frac{204}{100}$ or $2\frac{4}{100}$ **[M1]**

$2\frac{1}{25}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.375 **[B1]**

(c) 0.4 $\frac{5}{12}$ 42% $\frac{3}{7}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the fractions or percentages to decimals correctly, e.g., 0.416 . . . , 0.42, 0.428 . . .

[Back to Question] *[Total Marks: 4]*

6 **(a)** $\frac{75}{1000}$ or equivalent unsimplified fraction **[M1]**

$\frac{3}{40}$ **[A1]**

(b) 3.4 **[B1]**

(c) M1 for converting at least two of the other forms to decimals or percentages correctly (e.g. $\frac{5}{8} = 0.625$, $\frac{7}{11} = 0.636\ldots$, $62\% = 0.62$) **[M1]**

0.6 62% $\frac{5}{8}$ $\frac{7}{11}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

7 Length multiplier $= \frac{4}{3}$ **[B1]**

Area multiplier $= \frac{7}{6}$ **[B1]**

$\frac{4}{3} \times$ Width multiplier $= \frac{7}{6}$ oe **[M1]**

Width multiplier $= \frac{7}{6} \times \frac{3}{4} = \frac{7}{8}$ **[M1]**

$x = 12.5$ **[A1]**

[Back to Question] *[Total Marks: 5]*

8 (a) 60 **[B3]**

M1 for $60\% = \frac{3}{5}$ or $\frac{1}{5} = 20\%$ seen

M1 for $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$ or $60\% - 20\% = 40\%$

M1 for $\frac{24}{2} \times 5$ or $\frac{24}{40} \times 100$

(b) 400 **[B3]**

M1 for $100\% - 30\% = 70\%$ or $0.7x$ seen

M1 for $\frac{3}{4} \times 0.7x = 210$ or $0.525x = 210$ or equivalent equation

[Back to Question] *[Total Marks: 6]*

9 (a) Total number of counters $= x + (2x + 5) + (3x - 1) = 6x + 4$ **[M1]**

$\frac{2x+5}{6x+4} = \frac{35}{100}$ or $\frac{7}{20}$ **[M1]**

$20(2x + 5) = 7(6x + 4)$ leading to $40x + 100 = 42x + 28$ **[M1]**

36 **[A1]**

(b) Substituting $x = 36$ to find green counters $= 3(36) - 1 = 107$ and total counters $= 6(36) + 4 = 220$ **[M1]**

$\frac{107}{220} \times 100$ **[M1]**

$48\frac{7}{11}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** $1000x = 245.4545\ldots$ and $10x = 2.4545\ldots$ or $100x = 24.5454\ldots$ and $x = 0.24545\ldots$ **[M1]**

$990x = 243$ or $99x = 24.3$ **[M1]**

$\frac{27}{110}$ **[A1]**

(b) Identifies that the denominator $42 = 2 \times 3 \times 7$ (or states that 42 has prime factors other than 2 and 5) **[B1]**

Terminating decimals can only have prime factors of 2 and/or 5 in their denominator. Since the fraction is in its simplest form, the factors of 3 and 7 do not cancel out, so it must recur. **[B1]**

(c) $\frac{10x+y}{99} = \frac{k}{11}$ or $\frac{9k}{99}$ (or states that the two-digit number xy is a multiple of 9) **[M1]**

9 **[A1]**

[Back to Question] *[Total Marks: 7]*

11 **(a)** States appropriate pair of equations to eliminate the recurring part, e.g., $100x = 27.77\ldots$ and $10x = 2.77\ldots$ **[M1]**

Subtracts to get $90x = 25$ leading to $x = \frac{25}{90}$, and correctly simplifies this to $\frac{5}{18}$ **[A1]**

(b) **(i)** 36 **[B3]**

If 0 scored, allow [M1] for calculating the fraction or percentage of the whole school for either boys or girls, e.g., $\frac{2}{5} \times 45\%$ or $\frac{3}{5} \times 30\%$

Allow [M1] for evaluating either as 18%

(ii) 400 **[B2]**

If 0 scored, allow [M1] for a correct algebraic setup or division, e.g., $\frac{144}{0.36}$ or $36\% = 144$

[Back to Question] *[Total Marks: 7]*

12 **(a)** $x - \frac{1}{3}x + 12$ or $\frac{2}{3}x$ **[M1]**

$\frac{2}{3}x + 12$ **[A1]**

(b) Expression after 2nd stop: $\frac{3}{4}\left(\frac{2}{3}x + 12\right)$ or $0.75\left(\frac{2}{3}x + 12\right)$ **[M1]**

Expands correctly to $\frac{1}{2}x + 9$ **[M1]**

Equates to final condition: $\frac{1}{2}x + 9 = x - 15$ **[M1]**

48 **[A1]**

(c) Final number of passengers $= 48 - 15 = 33$ **[B1]**

$\frac{33}{48} \times 100$ or $\frac{11}{16} \times 100$ or $\frac{275}{4}$ **[M1]**

$68\frac{3}{4}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

13 (a) 315 **[B1]**

(b) 27.5 **[B2]**

M1 for $\frac{3}{8} \times 100 = 37.5\%$ or for $100 - (35 + 37.5)$

(c) $\frac{9}{160}$ **[B2]**

M1 for $\frac{15}{100} \times \frac{3}{8}$ or $\frac{47.25}{840}$

[Back to Question] *[Total Marks: 5]*

14 (a) Multiplier for 1st hour: $1 - 0.24 = 0.76$ (or 76%) **[B1]**

Multiplier for 2nd hour: $1 - \frac{2}{5} = \frac{3}{5} = 0.6$ (or 60%) **[B1]**

$0.76 \times 0.6 \times V = 45.6$ or $0.456V = 45.6$ **[M1]**

100 **[A1]**

(b) Total lost $= 100 - 45.6 = 54.4$ or $1 - 0.456 = 0.544$ **[M1]**

$\frac{68}{125}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

15 (a) $100\% - 45\% - 37.5\% = 17.5\%$ oe **[M1]**

$\frac{140}{0.175}$ oe **[M1]**

800 **[A1]**

(b) Junior students $= \frac{3}{8} \times 800 = 300$ **[M1]**

$1.5G + G = 300$ oe **[M1]**

120 **[A1]**

[Back to Question] *[Total Marks: 6]*

16 (a) Converts at least two of the non-decimals to decimals correctly (e.g. 0.71 and 0.75) **[M1]**

$71\%, 0.714, \frac{5}{7}, \frac{3}{4}$ **[A1]**

(b) $\frac{1}{2}\left(\frac{2}{5}+\frac{5}{8}\right)$ or equivalent **[M1]**

$\frac{41}{80}$ **[A1]**

(c) $9x - 4x = 35$ or states the multiplier is $35 \div 5 = 7$ **[M1]**

$\frac{28}{63}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

17 (a) $\frac{2}{5} = 40\%$ or 0.4, and total percentage of Sales and Production $= 40\% + 28\% = 68\%$ **[M1]**

$100\% - 68\% = 32\%$ and equating $0.32 \times$ Total $= 24$ or equivalent **[M1]**

75 **[A1]**

(b) Number of women in Sales $= 0.60 \times 30 = 18$ **[M1]**

Number of women in Production $= \frac{3}{7} \times 21 = 9$ (or men $= 12$, so women $= 9$) **[M1]**

Total women $= 18 + 9 + 12 = 39$ **[M1]**

52 **[A1]**

[Back to Question] *[Total Marks: 7]*

18 (a) 369 **[B2]**

If 0 scored, allow [M1] for 450×0.82 or 450×0.18

(b) 740 **[B3]**

If 0 scored, allow [M2] for $\frac{629}{1-0.15}$ or $\frac{629}{0.85}$

If 0 scored, allow [M1] for identifying that $85\% = \$629$

(c) 4 **[B2]**

If 0 scored, allow [M1] for a correct calculation showing the compound effect, e.g., $1.2 \times 0.8 = 0.96$ or choosing a starting value of 100 and reaching 96

[Back to Question] *[Total Marks: 7]*